AI-POWERED SUPPLY CHAIN DISRUPTION ASSISTANT

Design Document

System Architecture and Implementation Design

Project: AI-Powered Supply Chain Disruption Assistant

1. Document Purpose and Scope

This document describes the complete system design of the AI-Powered Supply Chain Disruption Assistant — a local decision-support application that compares new supply-chain disruption events against 100,000 historical case records. It covers functional and non-functional requirements, technology selection rationale, data pipeline, hybrid retrieval strategy, machine learning models, LangGraph orchestration, API design, database schema, frontend architecture, and evaluation workflow.

2. Problem Statement

Global supply chains face frequent disruptions from factory incidents, cyber attacks, port congestion, labor strikes, geopolitical events, and natural disasters. Operations and procurement teams urgently need empirical evidence from prior disruptions to triage new events. Relying on intuition leads to delayed responses, suboptimal mitigations, and elevated financial losses.

The system addresses this by: (1) retrieving historically comparable cases using hybrid semantic and keyword search, (2) predicting operational impacts using four trained Random Forest models, and (3) providing evidence-backed recommendations with explicit human-review safeguards when evidence is insufficient.

3. Business Objectives

- Accelerate triage — reduce disruption triage from hours to minutes by surfacing historical precedents instantly.
- Evidence-based mitigation — ground recommendations in empirical historical outcomes, not unconstrained generation.
- Quantify risk — predict production impact (%), revenue loss (USD), recovery days, and supplier change probability.
- What-If simulation — compare four mitigation strategies using historical statistics before committing resources.
- Human-in-the-loop safeguards — flag low-evidence scenarios for mandatory operator review.

4. Functional Requirements

ID	Requirement
FR-01	Accept disruption event: description, type, industry, tier, region, size, backup availability, severity.
FR-02	Retrieve top-K similar historical cases via ChromaDB + BM25 hybrid search with SQL metadata filters.
FR-03	Predict production impact (%), revenue loss (USD), full recovery days, and permanent supplier change probability.
FR-04	Execute deterministic 10-node LangGraph workflow: classify, investigate, retrieve, predict, response_analysis, recovery_analysis, resilience_analysis, priority, recommendation, validate.
FR-05	Generate evidence-backed recommendations; flag analyses with fewer than 3 precedents for human review.
FR-06	Support optional LLM (Gemini, Anthropic, OpenRouter); run fully offline with deterministic fallback.
FR-07	Persist analyses, predictions, recommendations, and evaluations to MySQL with user-level isolation.
FR-08	Provide paginated, filterable analysis history with full request/response data per authenticated user.
FR-09	Accept user feedback with actual outcome values; compute MAE drift against model predictions.
FR-10	Run RAGAS evaluation after every analysis without blocking the API response.
FR-11	Expose analytics: disruption counts by type, industry, region, severity; response medians; recovery histogram.
FR-12	Authenticate with scrypt hashing and 24-hour opaque bearer tokens; reject prompt-injection inputs.

5. Non-Functional Requirements

Category	Requirement
Performance	Initial retrieval < 5 seconds with indexes loaded; subsequent requests < 2 seconds.
Availability	All core analysis paths available offline without LLM API or external network access.
Security	Passwords as salted scrypt hashes; bearer tokens stored as SHA-256 hashes; no secrets in VCS.
Scalability	Dataset and vector index built offline; not reloaded per request.
Maintainability	Each component independently testable: retrieval, prediction, recommendation, evaluation.

6. System Components and Responsibilities

Component	File	Responsibility
API Gateway	backend/app/main.py	REST routes, startup lifecycle, auth dependencies, exception handlers, RAGAS invocation.
LangGraph Workflow	backend/app/agents/workflow.py	10-node sequential StateGraph; deterministic orchestration of all analysis stages.
Hybrid Retriever	backend/app/rag/hybrid_retriever.py	ChromaDB cosine (0.7 weight) + BM25 keyword (0.3 weight) fusion with SQL metadata filtering.
ML Predictor	backend/app/ml/predictor.py	Preprocessing, inference, TreeExplainer feature importance for 4 saved Random Forest models.
Analysis Service	backend/app/services/analysis.py	Historical strategy comparison, What-If simulation, recovery aggregation, resilience scoring.
Supplier Risk Service	backend/app/services/supplier_risk.py	Multi-factor supplier vulnerability indexing, regional exposure mapping, backup tracking.
Feedback Analytics	backend/app/services/feedback_analytics.py	MAE drift vs actual outcomes; automated retraining recommendations.
Security Module	backend/app/security.py	Scrypt hashing, opaque bearer token issuance, SHA-256 token storage, auth context.
Evaluation Service	backend/app/evaluation_service.py	RAGAS faithfulness, answer relevancy, context precision, and context recall computation.
Frontend SPA	frontend/src/	React 18 + Vite SPA with Recharts visualizations and Lucide icons.

7. Technology Stack and Justification

Component	Technology	Rationale
API Framework	FastAPI (Python 3.11+)	High performance, auto OpenAPI docs, Pydantic validation, async support.
Database	MySQL 8 + SQLAlchemy	ACID-compliant persistence for 100,000 records, users, history, and evaluations.
Vector Store	ChromaDB	Lightweight embedded on-disk collection with cosine similarity; no separate server.
Embedding Model	BAAI/bge-small-en-v1.5	Top MTEB compact class (384 dims); fast CPU inference; L2-normalized.
Lexical Search	rank-bm25	Exact-term matching complementing semantic search; captures domain jargon.
Orchestration	LangGraph	Deterministic named-node state machine; explicit inspection and error isolation.
Machine Learning	scikit-learn Random Forest	Robust tabular regression/classification; fast CPU inference; TreeExplainer support.
LLM Integration	Gemini / Anthropic / OpenRouter	Optional narrative synthesis; cascading fallback to deterministic extraction.
Frontend	React 18 + Vite	Hot Module Replacement; component modularity; modern UI responsiveness.
Visualizations	Recharts	SVG-based composable charts with animations and cross-browser consistency.

8. Data Pipeline and Processing

Dataset: backend/data/supply_chain_disruption_recovery.csv — 100,000 records, 15 columns.
External provenance not documented in repository artifacts.

Offline Pipeline

- inspect_dataset.py — profiles schema, distributions, missing values; writes evaluation/dataset_profile.json.
- ingest_dataset.py — validates schema, strips whitespace, enforces bounds (severity 1–5, impact 0–100%), deduplicates, bulk-inserts into MySQL.
- build_vector_index.py — renders each case as a textual document; generates L2-normalized BAAI/bge-small-en-v1.5 embeddings; indexes in ChromaDB with categorical metadata.
- build_bm25_index.py — tokenizes case text; serializes BM25 index to backend/rag/bm25_index.joblib.
- train_models.py — trains 4 Random Forest models on 70,000 rows; evaluates on 15,000 validation rows; saves .joblib artifacts and metrics to evaluation/model_metrics.json.

9. Data Preprocessing and Feature Engineering

Seven initial-event features used by all four prediction models:

Feature	Encoding	Notes
disruption_type	One-hot	6 categories: Port Congestion, Geopolitical, Labor Strike, Natural Disaster, Cyber Attack, Factory Incident.
industry	One-hot	5 categories: Electronics, Automotive, Consumer Goods, Pharmaceuticals, Aerospace.
supplier_tier	Ordinal integer 1–4	Passed as numeric; higher tier = less critical supplier.
supplier_region	One-hot	5 categories: Asia-Pacific, Europe, North America, South America, Africa/Middle East.
supplier_size	One-hot	Small, Medium, Large.
has_backup_supplier	Boolean (0/1)	Direct indicator of immediate mitigation capacity.
disruption_severity	Ordinal integer 1–5	Strongest individual predictor of impact and recovery.

Excluded (label leakage): production_impact_pct, revenue_loss_usd, full_recovery_days, permanent_supplier_change (targets); response_type, response_time_days, partial_recovery_days (post-event outcomes).

10. Machine Learning Models and Prediction Tasks

Task	Target	Type	Algorithm	Val Metric	Test MAE / F1
Production Impact	production_impact_pct	Regression	Random Forest	MAE	7.99 pp
Financial Loss	revenue_loss_usd	Regression	Random Forest	MAE	$1,774,595
Recovery Time	full_recovery_days	Regression	Random Forest	MAE	31.74 days
Supplier Change	permanent_supplier_change	Classification	Random Forest	F1	F1 = 0.401

Model selection: the training script compares Random Forest and Extra Trees by validation MAE (regression) or F1 (classification). SHAP TreeExplainer generates per-feature importance scores at inference time.

11. Hybrid Retrieval Strategy

Two-stage hybrid fusion: (1) ChromaDB cosine-distance candidates + BM25 keyword candidates in parallel. (2) SQL metadata filtering on disruption_type, industry, supplier_tier, supplier_region, has_backup_supplier, disruption_severity. (3) Score normalization: hybrid = 0.7 * semantic + 0.3 * bm25_normalized. (4) Top-K ranking by fused score; enrichment from MySQL structured outcomes.

Chunking: one complete case document per record; no splitting or overlap. Splitting would fracture correlated attributes (e.g., severity and full_recovery_days), destroying semantic coherence.

12. LangGraph Orchestration Workflow

Node	Responsibility
1. classify	Categorizes event severity, maps operational domains, determines triage level.
2. investigate	Constructs structured retrieval filters from event parameters.
3. retrieve	Executes hybrid Chroma/BM25 retrieval; gathers similar historical precedents.
4. predict	Invokes four ML models; generates impact estimates and feature importances.
5. response_analysis	Aggregates historical response types; compares outcome medians by strategy.
6. recovery_analysis	Calculates recovery distributions, medians, and IQR ranges.
7. resilience_analysis	Evaluates tier depth, backup availability, historic vulnerability; scores 0–100.
8. priority	Computes priority score from severity, predicted loss, and impact percentiles.
9. recommendation	Synthesizes evidence-backed strategy via LLM or deterministic rule engine.
10. validate	Asserts >= 3 precedents; flags failures for mandatory human review.

13. LLM Provider Routing and Fallback

- Primary — Google Gemini (LLM_PROVIDER=gemini, GEMINI_API_KEY).
- Secondary — Anthropic Claude (LLM_PROVIDER=anthropic, ANTHROPIC_API_KEY).
- Tertiary — OpenRouter via OpenAI-compatible API.
- Offline fallback — Deterministic historical extraction (LLM_PROVIDER=demo); zero network calls.

provider_tracking in every response records the final provider used and whether a fallback occurred.

14. API Design

Route Group	Endpoints
/api/auth/*	POST /register, POST /login, POST /logout, GET /me
/api/health	GET — database, Chroma, BM25, LangGraph, model status
/api/disruptions/*	GET /disruptions (paginated), GET /{id}, POST /analyze (full workflow)
/api/retrieval/similar	POST — standalone hybrid search with filters
/api/predictions	POST — direct ML prediction endpoint
/api/analysis/what-if	POST — four-strategy What-If simulation
/api/recommendations	POST — retrieve stored recommendation by analysis_id
/api/suppliers/risk	GET — supplier vulnerability and regional risk intelligence
/api/history/*	GET (paginated/filtered), GET /{id}, DELETE /{id}, DELETE (clear all), GET /filters, GET /stats
/api/analytics/*	GET /overview, GET /disruptions, GET /responses, GET /recovery
/api/feedback/*	POST (submit), GET (analysis + drift), GET /stats
/api/evaluation/*	GET (model metrics + RAGAS logs), GET /test-cases (deprecated)
/api/models	GET — model availability and training metrics

15. Database Design

Table	Purpose
users	Accounts: id, name, email, password_hash (scrypt), created_at, is_active.
auth_sessions	Active sessions: token_hash (SHA-256), user_id, created_at, expires_at.
disruptions	100,000 ingested case records with all 15 schema fields plus auto-increment id.
analysis_requests	JSON request payload; parent record for history/prediction/recommendation.
analysis_history	Per-user analysis records: disruption_type, industry, region, tier, severity, provider, fallback_used, response_time_ms, status, search_text.
recommendations	JSON recommendation payloads linked to analysis_requests.
predictions	JSON prediction results and feature importances linked to analysis_requests.
feedback	User outcomes: useful, actual_response, actual_recovery_days, actual_production_impact_pct, actual_revenue_loss_usd, comment.
evaluation_results	RAGAS and evaluation JSON results linked to analysis_requests.
model_versions	Metadata for tracking trained artifact versions.

16. Frontend Design

React 18 SPA built with Vite. Communicates exclusively with FastAPI via Axios. Key pages:

- Overview — summary KPIs, priority disruption list, interactive disruption table.
- Disruption Analysis — event input form, result panels: classification, similar cases, predictions, recommendation, RAGAS evaluation.
- History — paginated, filterable user history with full request/response expansion.
- Similar Cases — standalone hybrid search interface.
- Predictions — standalone ML prediction form with feature importance visualization.
- What-If / Recommendations — four-strategy comparative simulation.
- Supplier Risk — risk dashboards by industry, tier, region.
- Risk and Resilience Profiling — resilience score analysis with event profile display.
- Analytics — Recharts visualizations: distribution, response medians, recovery histogram.
- Feedback — outcome submission form linked to completed analyses.
- Evaluation — RAGAS metrics viewer, model performance summary, benchmark results.

17. Security and Access Control

- Password storage: scrypt key derivation with individual cryptographic salts.
- Session tokens: cryptographically random 64-character opaque bearer tokens; stored as SHA-256 hashes; expire after 24 hours.
- User isolation: all analyses, history, and feedback partitioned by authenticated user_id.
- Input guardrails: description field (12–1200 chars) validated by regex rejecting prompt-injection patterns.
- Secret management: API keys and DB credentials in root .env; excluded from VCS via .gitignore.

18. Error Handling and Fallback Resilience

Error Code	Condition	Handler
HTTP 422	Pydantic validation failure	Field-level error detail array.
HTTP 404	Non-existent or unowned resource	Descriptive error message.
HTTP 401	Invalid or expired bearer token	WWW-Authenticate header.
HTTP 409	Duplicate email on registration	Descriptive conflict message.
HTTP 503	Missing .joblib models, indexes, or LangGraph; DB unreachable	Controlled JSON with error detail.
HTTP 500	Unexpected server error	Generic message; traceback suppressed from response.

LLM fallback chain: Gemini → Anthropic → OpenRouter → Deterministic historical extraction.
Startup index loading is deferred to a background daemon thread to avoid blocking API startup.

19. Design Trade-offs

- Fixed hybrid weights (0.7/0.3) vs learned re-ranker — eliminates cold-start overhead; accepts marginally lower ranking flexibility.
- Tabular ChromaDB embeddings vs graph database — simpler deployment; avoids Neo4j operational complexity.
- Random Forest vs deep neural networks — favors CPU inference speed, tabular interpretability, and native TreeExplainer support.
- One document per case vs sub-case chunking — preserves attribute coherence at the cost of longer documents.

20. Limitations

- Dataset provenance is undocumented — the repository does not state whether records are synthetic, operational, or otherwise derived.
- Revenue loss (R2 0.374) and recovery time (R2 0.369) predictions have moderate accuracy; unsuitable for exact financial planning.
- Resilience score is a domain-weighted heuristic, not an ISO-standardized metric.
- No temporal timestamps or supplier identifiers in the dataset; temporal trend and entity-level analysis are not supported.

21. Conclusion

The AI-Powered Supply Chain Disruption Assistant delivers a complete, locally operable end-to-end disruption triage system combining hybrid retrieval, four Random Forest prediction models, a deterministic 10-node LangGraph workflow, optional multi-tier LLM synthesis, RAGAS quality evaluation, and a modern React/Vite dashboard — all fully operational without external network access.